DRY RUN TEST OUTPUTS

UNIT 1 ASSESSMENT TOOL 2

Question 1: Dry Run of A Search Algorithm

Normal output :

```
Current Node: A
g(n): 0
h(n): 7
f(n): 7

Current Node: B
g(n): 2
h(n): 6
f(n): 8

Current Node: E
g(n): 4
h(n): 2
f(n): 6

Current Node: G
g(n): 6
h(n): 0
f(n): 6

Optimal Path: A -> B -> E -> G
Total Cost: 6
```

When input changed from A – B and also the Goal G – D

```
Current Node: B
g(n): 0
h(n): 6
f(n): 6

Current Node: E
g(n): 2
h(n): 2
f(n): 4

Current Node: G
g(n): 4
h(n): 0
f(n): 4

Current Node: C
g(n): 3
h(n): 4
f(n): 7

Current Node: D
g(n): 7
h(n): 3
f(n): 10

Optimal Path: B -> D
Total Cost: 7
>
```

Input = initial A and Goal state changed from G – D

```
Current Node: A
g(n): 0
h(n): 7
f(n): 7

Current Node: B
g(n): 2
h(n): 6
f(n): 8

Current Node: E
g(n): 4
h(n): 2
f(n): 6

Current Node: G
g(n): 6
h(n): 0
f(n): 6

Current Node: C
g(n): 4
h(n): 4
f(n): 8

Current Node: D
g(n): 9
h(n): 3
f(n): 12

Optimal Path: A -> B -> D
Total Cost: 9
```

Question 2: Dry Run of Minimax with Alpha-Beta Pruning

Normal output :

```
1.py
Enter 3 leaf values for Left MIN (space separated): 3 5 6
Enter 3 leaf values for Right MIN (space separated): 9 1 2

----- Left MIN Node -----
Evaluating: 3
Alpha = -inf  Beta = 3
Evaluating: 5
Alpha = -inf  Beta = 3
Evaluating: 6
Alpha = -inf  Beta = 3
Selected Value of Left MIN = 3

----- Right MIN Node -----
Evaluating: 9
Alpha = 3  Beta = 9
Evaluating: 1
Alpha = 3  Beta = 1
Selected Value of Right MIN = 1

----- Final Result -----
Final Minimax Value = 3
Best Move for MAX = Left Subtree
Pruned Node(s): [2]
```

Input changed output :

```
Enter 4  leaf values for Left MIN (space separated): 1 2 3 4
Enter 4  leaf values for Right MIN (space separated): 5 6 7 8

----- Left MIN Node -----
Evaluating: 1
Alpha = -inf  Beta = 1
Evaluating: 2
Alpha = -inf  Beta = 1
Evaluating: 3
Alpha = -inf  Beta = 1
Evaluating: 4
Alpha = -inf  Beta = 1
Selected Value of Left MIN = 1

----- Right MIN Node -----
Evaluating: 5
Alpha = 1  Beta = 5
Evaluating: 6
Alpha = 1  Beta = 5
Evaluating: 7
Alpha = 1  Beta = 5
Evaluating: 8
Alpha = 1  Beta = 5
Selected Value of Right MIN = 5

----- Final Result -----
Final Minimax Value = 5
Best Move for MAX = Right Subtree
No Nodes Pruned
```